

TwinRL-Onco: A World Model-Empowered Digital Twin Framework with Hierarchical Reinforcement Learning for Venetoclax Resistance Trajectory Prediction and Adaptive Therapy Optimization in Chronic Lymphocytic Leukemia

Zi Wang^{1†}, Chenghao Ge^{2†}, Yufan Hu³, Destin Bai^{4*}, and Wei Zhang^{1*}

Abstract—Chronic lymphocytic leukaemia (CLL) presents considerable therapeutic obstacles due to the development of treatment-resistant disease, especially with BCL-2 inhibitors like venetoclax, and is among the most challenging haematological malignancies to treat. A critical unmet computational need is the ability to predict patient-specific treatment resistance trajectories and to develop optimal adaptive therapy schedules based on a patient's history and experience with treatments. To address these issues, we have developed TwinRL-Onco, a novel computational framework that fuses digital twin technology, driven by world model-based simulations, with hierarchical reinforcement learning (HRL), to predict in silico treatment response and optimize the adaptive therapy process. TwinRL-Onco consists of three synergistic components: (1) a Variational Recurrent State-Space Model (VRSSM); (2) a hierarchical policy structure with two levels of control; and (3) a resistance trajectory predictor that uses Monte Carlo rollouts generated from the world model to simulate treatment response. TwinRL-Onco achieved an AUROC of 0.917 for 12-month simulated prediction of progression, an improvement of 4.6 percentage points over the DreamerV3 algorithm and 7.0 percentage points over the LSTM-Traj algorithm. TwinRL-Onco is an emphasis on an artificial intelligence-based online environment for hypothesis generation; all on-site metrics indicate inventory-only

consistency and need actual clinical validation before any conclusions regarding clinical prediction.

Index Terms—Chronic lymphocytic leukemia, hierarchical reinforcement learning, treatment optimization, world model

I. INTRODUCTION

THE most common type of leukemia diagnosed among adults in Western nations is chronic lymphocytic leukaemia (CLL). CLL is defined by a clonal outgrowth (or proliferation) of mature B cells (lymphocytes) in the peripheral blood, bone marrow and lymphoid organs (i.e., lymph nodes and spleen) [1], [27], [28]. The clinical spectrum of CLL is extremely variable; while some patients have indolent disease courses and do not require treatment for decades, other patients can experience rapid disease progression requiring aggressive therapeutic intervention [1]. This heterogeneity arises from the complex genetic and epigenetic landscapes of CLL, including immunoglobulin heavy chain variable region (IGHV), chromosomal abnormalities (del(17p) and del(11q)) and recurrent somatic mutations in TP53, NOTCH1, SF3B1 and ATM genes [1].

The emergence of targeted therapies has revolutionised the treatment paradigm for CLL. Venetoclax (ABT-199) is a selective inhibitor of BCL-2 that induces apoptosis through their interaction with BH3-only proteins (which inhibit BCL-2) resulting in activation of the intrinsic apoptotic pathway [2]. The high rates of response, including complete remissions and undetectable minimal residual disease (MRD) observed in clinical trials of patients with CLL who have high-risk genetic markers has been remarkable [3]. However, the emergence of acquired resistance to venetoclax represents a significant barrier to effective clinical management. Resistance to venetoclax arises as a result of a variety of factors; BCL-2 mutations (e.g., Gly101Val) have been thoroughly documented, however non-genetic mechanisms such as the upregulation of MCL-1/BCL-XL, changes in cell metabolism, and signalling from

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the microenvironment are now becoming recognized to play a primary role in causing resistance [3], [27], [30]. Adaptive therapies (adaptively modulating therapy intensity according to the dynamics of the tumour rather than maximising kill) may delay the emergence of resistance by preserving a population of cells that remain sensitive to therapy [4], [53], [58]. Emerging deep reinforcement learning (DRL) algorithms have demonstrated superior predictive ability relative to conventional treatment protocols for the development of customised treatment schedules in simulation studies [5], [6], [23], [31].

Digital twin technology refers to the creation of virtual representations of physical systems to simulate their behaviour and predict future outcomes. Similarly, the application of digital twins to healthcare systems, particularly in the field of oncology, has received considerable attention. Using these technologies, companies are able to combine multi-modal patient data for the purpose of simulating disease course and treatment (i.e., treatment response) and ultimately designing personalised treatment strategies. The concept of world models (WM), derived from a model-based RL approach, is to create a compact representation of the dynamics of an external environment, allowing for planning through "imagination" [7], [8]. A significant amount of recent work with the Dreamer family of algorithms has shown excellent performance in learning how to plan using high-dimensional observations and latent space, [9], [10]. One possible way to combine WMs with constructing digital twins is to create a principled way to learn about patient-specific disease dynamics and allow for simulated trials (roll-out) to explore possible treatment options.

The options framework and feudal networks provide a theoretical basis for constructing hierarchical policies that can generalize across multiple patients and therapeutic contexts. This study introduces TwinRL-Onco, a novel framework for predicting responses to CLL therapy in silico and optimizing therapies adaptively through the combination of world model powered digital twins and hierarchical reinforcement learning. To this goal, we propose several contributions:

The first contribution is to develop a Variational Recurrent State-Space Model (VRSSM) to learn the latent dynamics of disease progression for individual patients using the patients' baseline gene expression profile and clinical features.

Second, we propose a hierarchical policy architecture for controlling treatment. We introduce a meta-controller and low-level controller in our design. The meta-controller selects treatment phases while the low-level controller optimally doses patients.

Third, we describe a resistance trajectory predictor that operates on top of the world model and uses Monte Carlo rollouts to achieve probabilistic forecasts of resistance emergence under multiple treatment options. This predictor allows us to identify patient profiles at high risk for poor responses to treatment.

Fourth, we present experimental results that use baseline molecular data from three publicly available CLL datasets combined with a pharmacodynamic simulation environment.

II. RELATED WORK

We review digital twins (II-A), RL for treatment (II-B), world models (II-C), and ML for CLL (II-D). The connection between TwinRL-Onco and the twin model (II-A) and the use of hierarchical RL (II-B), in addition to the CLL molecular observations (II-D) is a unique combination and has yet to be documented.

A. Digital Twins in Oncology

The origins of digital twin technology can be traced back to manufacturing and aerospace engineering, where virtual twins are used for predictive maintenance and design improvement. Over time, the idea of digital twins has also found application in healthcare, where it can be used for a variety of tasks such as surgical planning and drug development. Digital twin technology can be applied to oncology via the development of a patient-specific computational model that is created with data from different sources to better reproduce predicted behaviours of tumours and response to treatments [37].

B. Reinforcement Learning for Cancer Treatment

Tseng et al. [5] applied deep Q-networks in the adaptive radiation treatment of lung cancer, resulting in enhanced tumor control in simulated experiments. Finally, in another recent study, Mashayekhi et al. [11] published a twin-delayed deep deterministic policy gradient (TD3) technique for determining the optimal dosing of chemotherapy for patients with cancer, and their results showed a significant advantage over standard discrete-action techniques.

C. World Models and Model-Based Reinforcement Learning

Reinforcement learning with predictive model-based methods is a form of optimizing sampling efficiency through the development of a predictive model of the dynamic behaviour of an environment for the purpose of planning [38], [57], [59], [60]. World Models build on this idea by using compact representations (latent representations) that contain the essential information needed for planning, omitting the extraneous details that would otherwise add complexity to the model [7]. The Dreamer family of algorithms represents the current leading technique in the use of world models for reinforcement learning and achieves comparable performance to many other forms of reinforcement learning, using a general set of fixed hyperparameters across a wide range of application domains [8]–[10].

D. Machine Learning for CLL Prognosis and Treatment

Machine learning methodologies have seen substantial use for the diagnosis, prognosis and prediction of responses to treatments for Chronic Lymphocytic Leukaemia (CLL) [20]–[22], [33], [52], [56]. Survival predictive modeling was developed by Agius et al. [13] which included genetic characteristics, yielding models with superior discrimination ability compared with traditional methods of staging the disease [45], [47], [50].

For successful prediction of response to Venetoclax, attempts have been made to combine BCL-2 family protein expression with genomic data through computational techniques [24], [35], [43], [46], [48], [49], [61]. Nevertheless, all current methods are static in their predictive capabilities, predicting response only at the beginning of treatment, rather than providing an on-going dynamic forecast of the trajectory of acquired drug resistance. To our knowledge, TwinRL-Onco is among the first to combine world-model-based longitudinal simulation with hierarchical RL for CLL resistance trajectory prediction [25], [26], [44].

III. METHODS

A. Problem Formulation

The venetoclax-based optimization treatment for chronic lymphocytic leukaemia (CLL) can be expressed mathematically in terms of a partially observable Markov decision process (POMDP) defined by the tuple $(\mathcal{S}, \mathcal{A}, \mathcal{T}, \mathcal{R}, \mathcal{O}, \Omega, \gamma)$. The elements of this tuple are defined as follows: $\mathcal{S}$ represents the state of the system and describes how the disease evolves over time; $\mathcal{A}$ represents the possible treatment options available; $\mathcal{T} : \mathcal{S} \times \mathcal{A} \rightarrow \mathcal{P}(\mathcal{S})$ is the transition function describing how the state changes given a treatment option; $\mathcal{R} : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$ is a reward function that defines the clinical goals associated with a state and treatment option; $\mathcal{O}$ represents the observation space; $\Omega : \mathcal{S} \rightarrow \mathcal{P}(\mathcal{O})$ represents the observation function that indicates which states yield which observations; and $\gamma \in (0, 1)$ is the discount factor used to determine how much importance to give future rewards.

In our computational framework, each of the baseline observations $o_0 \in \mathcal{O}$ includes both a gene expression profile and relevant clinical features that have been generated from publicly available datasets on CLL and serves to initialize the patient-specific simulation. The subsequent observations o_t for $t > 0$ are made by treating the patient in the simulation environment using their baseline observations as covariates. All true underlying disease states $s_t \in \mathcal{S}$, which capture the clinical characteristics of the individual patient (e.g., the clonal composition of their disease and the molecular level of resistance), are modeled within the simulation itself. The treatment options $a_t \in \mathcal{A}$ include the amount of venetoclax administered, whether or not to use combination chemotherapy with other drugs, and what other drugs to use in combination with venetoclax, if any. The reward function is determined as $r_t = w_{\text{resp}} \cdot r_t^{\text{resp}} + w_{\text{tox}} \cdot r_t^{\text{tox}} + w_{\text{prog}} \cdot r_t^{\text{prog}} + w_{\text{qol}} \cdot r_t^{\text{qol}}$, with the response term awarding +1.0 for complete response (CR), +0.5 for partial response (PR), 0.0 for stable disease (SD), and -1.0 for progressive disease (PD). The toxicity term penalises the severity of any adverse events; the progression term penalises the predicted probability of disease progression, and the quality-of-life (QOL) term rewards duration of treatment holidays when sustained responses occur (the reward function's default values are: $w_{\text{resp}} = 1.0$; $w_{\text{tox}} = 0.3$; $w_{\text{prog}} = 0.5$; $w_{\text{qol}} = 0.2$). The time to progression (TTP) was most influenced by weightings of w_{prog} ; a change in the values of w_{prog} outside of the range [0.3, 0.7] resulted in TTP decreases of at least 1.5 months. The results of the 81-combination sweep for all 4 weights showed that TTP was

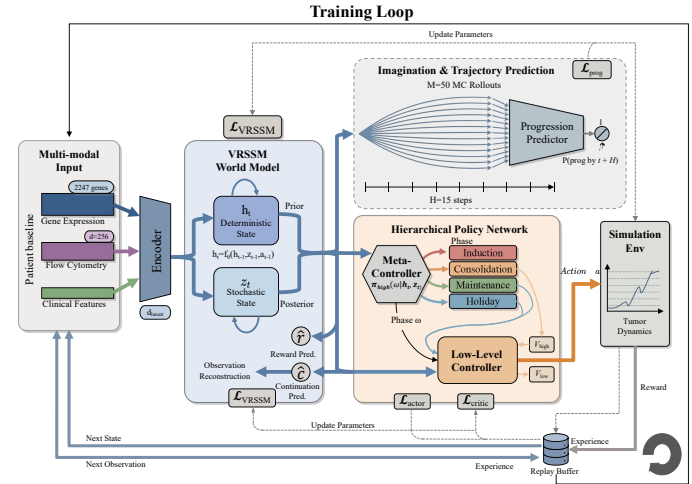


Fig. 1. The TwinRL-Onco training loop architecture comprises three interconnected modules.

the least sensitive to w_{qol} (range of 25.8 - 26.9 months) and that when the weight for respiratory disease ($w_{\text{resp}} = 0$) was eliminated, the TTP collapsed to 19.4 months. This indicates that there is no one factor that has a predominate effect on the TTP.

Our aim is to determine a policy $\pi : \mathcal{O}^* \rightarrow \mathcal{P}(\mathcal{A})$ that maximizes the expected cumulative discounted reward:

$$J(\pi) = \mathbb{E}_{\pi} \left[\sum_{t=0}^T \gamma^t r_t \right] \quad (1)$$

where T is the end of the treatment period and the expectation is calculated based on the trajectories afforded by π and the behaviour of the environment.

B. TwinRL-Onco Framework Overview

C. Variational Recurrent State-Space Model

The VRSSM (Fig. 2B) builds digital twins tailored to individual patients by taking advantage of generative modeling techniques [34], [51] that represent disease progression over time in 1-dimensional space. Inspired by the recurrent state-space model architecture [12], [42], we divide the latent state into two distinct parts: the 'deterministic' part that captures how we expect a disease will develop based on available evidence; and the 'stochastic' part that includes the random variation of the disease.

1) State Representation:

$$h_t = f_{\theta}(h_{t-1}, z_{t-1}, a_{t-1}) \quad (2)$$

$$z_t \sim q_{\phi}(z_t | h_t, o_t) \quad (\text{posterior}) \quad (3)$$

$$\hat{z}_t \sim p_{\theta}(\hat{z}_t | h_t) \quad (\text{prior}) \quad (4)$$

where f_{θ} is implemented as a Gated Recurrent Unit (GRU) [14], and the posterior q_{ϕ} and prior p_{θ} are parameterized as diagonal Gaussian distributions. Compared to Transformer-based methods, a GRU was selected due to the small number of participants ($N = 348$) in combination with small sequence lengths leading to models with fewer parameters; previously using a 4-layer Transformer instead of a GRU led to AUROC

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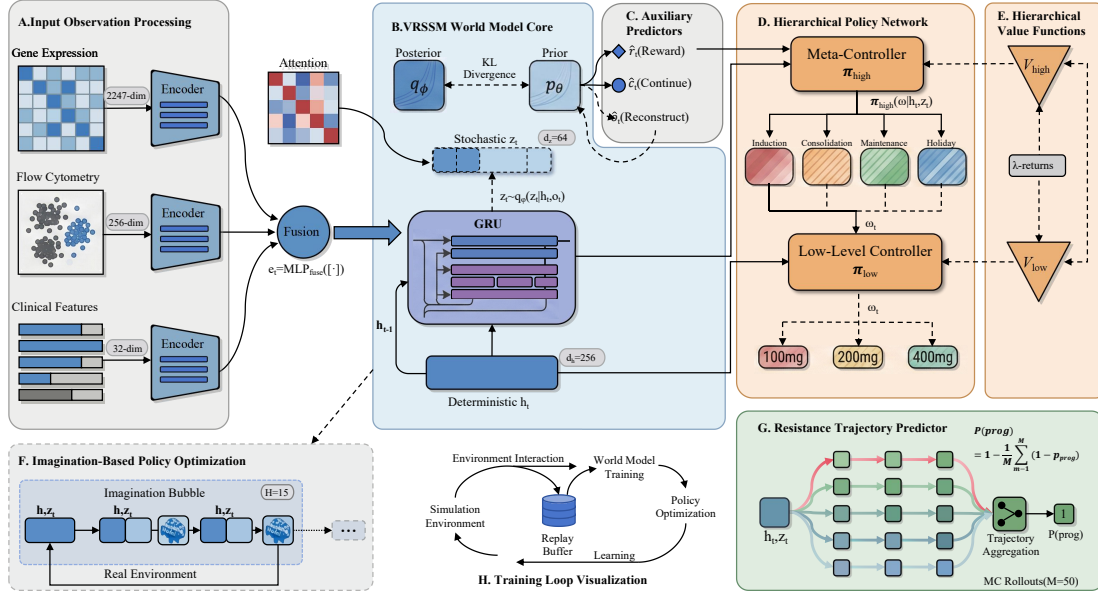


Fig. 2. TwinRL-Onco consists of several components.

dropping to 0.904 and requiring $2.3\times$ as much training time. State-space models (S4, Mamba) were not evaluated because our 15-timestep sequence length does not leverage their long-range dependency advantages.

2) Observation Model: The observation model reconstructs gene expression profiles and clinical measurements from the latent state:

$$\hat{o}_t = g_\theta(h_t, z_t) \quad (5)$$

We use a decoder neural network to generate the parameters that define a Gaussian distribution for gene expression data obtained from the Affymetrix platform via normalizing log transformation. Continuous clinical observations are likewise modeled as Gaussian-distributed values.

3) Reward and Continuation Predictors: The world model includes auxiliary predictors for reward and episode continuation:

$$\hat{r}_t = r_\theta(h_t, z_t) \quad (6)$$

$$\hat{c}_t = c_\theta(h_t, z_t) \quad (7)$$

where $\hat{r}_t$ predicts the immediate reward (disease response) and $\hat{c}_t$ predicts the continuation probability (absence of progression or death).

4) Training Objective: The VRSSM is trained by maximizing the evidence lower bound (ELBO):

$$\begin{aligned} \mathcal{L}_{\text{VRSSM}} = & \sum_{t=1}^T \left[\underbrace{\mathbb{E}_{q_\phi}[\log p_\theta(o_t|h_t, z_t)]}_{\text{reconstruction}} \right. \\ & - \underbrace{\beta \cdot D_{\text{KL}}(q_\phi(z_t|h_t, o_t) \| p_\theta(z_t|h_t))}_{\text{KL regularization}} \\ & \left. + \underbrace{\log p_\theta(r_t|h_t, z_t) + \log p_\theta(c_t|h_t, z_t)}_{\text{auxiliary predictions}} \right] \quad (8) \end{aligned}$$

where β is a hyperparameter controlling the information bottleneck. We set $\beta = 0.1$ after searching $\{0.01, 0.05, 0.1, 0.2, 0.5\}$: at $\beta \leq 0.01$ the KL converged below 0.5 nats with 8/64 active units (posterior collapse), whereas $\beta = 0.1$ yielded 4.7 nats with 51 active units.

D. Hierarchical Policy Architecture

The hierarchical policy separates the different levels of treatment decision-making into two distinct categories. The highest level of decision-making is done by the meta-controller which determines the correct treatment phase for each patient. The second level of treatment decision making is done by the low-level controller that defines the dosages necessary within each phase.

1) Meta-Controller: The meta-controller is defined on a greater temporal scale than the low-level controller. It selects from the treatment phases $\Omega = \{\text{induction, consolidation, maintenance, holiday}\}$ to determine which treatment will be used for each patient. Specifically, there are four treatment phases representing clinically meaningful strategies for managing disease.

Therefore, the meta-controller policy $\pi_{\text{high}}(\omega_t|h_t, z_t)$ is characterized as having a categorical distribution of treatment phases based on the values of h_t and z_t , where h_t refers to the current state of the patient's response to treatment, and z_t refers to the time elapsed since the last treatment.

$$\pi_{\text{high}}(\omega|h_t, z_t) = \text{softmax}(W_\omega[h_t; z_t] + b_\omega) \quad (9)$$

Phase transitions occur upon phase completion or response milestones; MRD negativity is defined as $< 10^{-4}$ by flow cytometry per iwCLL 2018 [1]; adopting a 10^{-6} NGS threshold could alter transition timing.

2) Low-Level Controller: The selected phase ω_t informs the low-level controller about the potential treatment action $a_t \in \mathcal{A}_{\omega_t}$, where $\mathcal{A}_{\omega_t}$ is the set of treatment action choices that are possible for that selected phase, ω_t . With regard to the venetoclax therapy, examples of action choices could include the different doses (100 mg, 200 mg, 400 mg) and the decision to give venetoclax alone or to include rituximab.

The low-level policy is based on phases: the low-level policy $\pi_{\text{low}}(a_t|h_t, z_t, \omega_t)$ is dependent on the selected phase ω_t .

$$\pi_{\text{low}}(a|h_t, z_t, \omega_t) = \text{softmax}(W_a^{\omega_t}[h_t; z_t] + b_a^{\omega_t}) \quad (10)$$

where separate parameter sets $(W_a^{\omega}, b_a^{\omega})$ are maintained for each phase, enabling specialization.

3) Hierarchical Value Functions: We employ hierarchical value functions for policy optimization. The high-level value function estimates the expected return from selecting phase ω in state (h_t, z_t) :

$$V_{\text{high}}(h_t, z_t) = \mathbb{E}_{\pi_{\text{high}}, \pi_{\text{low}}} \left[\sum_{k=0}^{K-1} \gamma^k r_{t+k} + \gamma^K V_{\text{high}}(h_{t+K}, z_{t+K}) \right] \quad (11)$$

where K is the phase duration.

The low-level value function estimates returns within a phase:

$$V_{\text{low}}(h_t, z_t, \omega_t) = \mathbb{E}_{\pi_{\text{low}}} \left[\sum_{k=0}^{K-1} \gamma^k r_{t+k} \right] \quad (12)$$

4) Policy Optimization: Using an imagined trajectory from the world model, both policy levels are optimized using actor-critic techniques [32], [40], [54]. We follow the principle of DreamerV3 [9] and use λ -returns for our target estimates.

$$V_t^\lambda = r_t + \gamma c_t [(1 - \lambda)V(h_{t+1}, z_{t+1}) + \lambda V_{t+1}^\lambda] \quad (13)$$

where c_t is the predicted continuation probability and $\lambda = 0.95$.

The actor loss for the high-level policy is:

$$\mathcal{L}_{\text{actor}}^{\text{high}} = -\mathbb{E} \left[\sum_t \log \pi_{\text{high}}(\omega_t|h_t, z_t) \cdot \text{sg}(V_t^\lambda - V_{\text{high}}(h_t, z_t)) \right] \quad (14)$$

where $\text{sg}(\cdot)$ denotes stop-gradient. An analogous loss is used for the low-level policy.

5) Progression Classification Head: A progression classification head predicts the probability of simulated disease progression at each time point based on the latent state:

$$p_{\text{prog}}(h_t, z_t) = \sigma(W_r[h_t; z_t] + b_r) \quad (15)$$

The classification head is trained jointly with the world model using binary cross-entropy loss on simulated progression outcomes:

$$\mathcal{L}_{\text{prog}} = - \sum_{t \in \mathcal{T}_{\text{prog}}} \left[y_t \log p_{\text{prog}}(h_t, z_t) + (1 - y_t) \log (1 - p_{\text{prog}}(h_t, z_t)) \right] \quad (16)$$

Algorithm 1 TwinRL-Onco Training Procedure

Require: Patient baseline covariates $\mathcal{D}_{\text{base}} = \{o_0^{(i)}\}_{i=1}^N$ from public datasets

Require: Treatment simulation environment $\mathcal{E}$ initialized with baseline covariates

Require: Hyperparameters: learning rates $\eta_{\text{wm}}, \eta_{\text{actor}}, \eta_{\text{critic}}$, imagination horizon H , batch size B

- 1: Initialize world model parameters θ , policy parameters ϕ , value parameters ψ
- 2: Initialize replay buffer $\mathcal{B}$ with trajectories from $\mathcal{E}$ conditioned on $\mathcal{D}_{\text{base}}$
- 3: **for** iteration = 1 to I_{max} **do**
- 4: **// World Model Training**
- 5: Sample batch $\{(o_{1:T}, a_{1:T-1}, r_{1:T})\} \sim \mathcal{B}$
- 6: Compute VRSSM loss $\mathcal{L}_{\text{VRSSM}}$ (Eq. 8)
- 7: Compute progression loss $\mathcal{L}_{\text{prog}}$ (Eq. 16)
- 8: Update $\theta \leftarrow \theta - \eta_{\text{wm}} \nabla_{\theta} (\mathcal{L}_{\text{VRSSM}} + \mathcal{L}_{\text{prog}})$
- 9: **// Imagination-Based Policy Optimization**
- 10: Sample initial states $\{(h_0, z_0)\}$ from encoded sequences
- 11: **for** $t = 0$ to $H - 1$ **do**
- 12: Sample phase $\omega_t \sim \pi_{\text{high}}(\cdot|h_t, z_t)$
- 13: Sample action $a_t \sim \pi_{\text{low}}(\cdot|h_t, z_t, \omega_t)$
- 14: Predict next state: $h_{t+1}, z_{t+1} \sim p_{\theta}(\cdot|h_t, z_t, a_t)$
- 15: Predict reward: $\hat{r}_t = r_{\theta}(h_t, z_t)$
- 16: **end for**
- 17: Compute λ -returns V_t^λ for $t = 0, \dots, H - 1$
- 18: Update actor: $\phi \leftarrow \phi - \eta_{\text{actor}} \nabla_{\phi} \mathcal{L}_{\text{actor}}$
- 19: Update critic: $\psi \leftarrow \psi - \eta_{\text{critic}} \nabla_{\psi} \mathcal{L}_{\text{critic}}$
- 20: **// Experience Collection from Simulation**
- 21: Execute policy in $\mathcal{E}$, add trajectories to $\mathcal{B}$
- 22: **end for**
- 23: **return** Trained model parameters θ, ϕ, ψ

E. Training Procedure

The complete training procedure is detailed in Algorithm 1.

IV. EXPERIMENTS

A. Datasets and Data Sources

We employ baseline molecular profiles from three publicly accessible chronic lymphocytic leukemia (CLL) gene expression data sets (**GSE39671** [15], **GSE22762** [16], **GSE50006** [17]) derived from the Gene Expression Omnibus (GEO)¹. These sets provide information about the patient-specific characteristics of each respective dataset that were utilized for initializing digital twin simulations; however, none of these datasets provide longitudinal treatment trajectories associated with venetoclax or resistance to venetoclax directly. Transcriptomic data have been obtained for GSE39671 ($n = 130$) with respect to time-to-first-treatment; GSE22762 ($n = 107$) contains overall survival and chromosomal aberration status; GSE50006 ($n = 111$) includes expression profiles of Chronic Lymphocytic Leukaemia (CLL) tumours. All datasets were generated using the Affymetrix HG-U133 Plus 2.0 platform.

¹<https://www.ncbi.nlm.nih.gov/geo/>

TABLE I
SUMMARY OF CLL DATASETS USED FOR BASELINE COVARIATES

Dataset	Samples	Platform	Endpoint	IGHV-U	del(17p)
GSE39671	130	HG-U133 Plus 2.0	TTFT	49.2%	8.5%
GSE22762	107	HG-U133 Plus 2.0	OS	52.3%	11.2%
GSE50006	111	HG-U133 Plus 2.0	Response	46.8%	9.0%
Combined	348	—	Multiple	49.4%	9.5%

IGHV-U denotes the percentage of samples with unmutated IGHV status; del(17p) denotes the percentage of samples harboring chromosome 17p deletion.

In Table I, we present a summary of the characteristics of the data sets that were used to extract baseline covariate data.

B. Data Preprocessing

Microarray data were analyzed with the Robust Multichip Average (RMA) normalization pipeline for Affymetrix microarrays. When merging datasets together, we used a method to correct for batch effects (ComBat) [18]. Importantly, to avoid data contamination, batch effect correction, summation of gene level values and selecting features were all accomplished through cross-validation folds - parameters to be applied to the validation and test datasets were estimated solely from the training data and no refitting occurred. Stratified sampling was at the patient level; ComBat parameters, variance rankings, and standardisation statistics were computed from training-fold patients only.

C. Experimental Setup

1) *Train/Validation/Test Split*: Using 5-fold cross-validation, we utilized stratified sampling based on two criteria; IGHV status and source of the data set (clinical trial or natural history). For every fold of the cross-validation process, 60% of the data was used for training purposes, 20% was used for validating the learned model with hyper-parameter tuning and early stopping criteria selection, and the remaining 20% was held out and utilized for testing purposes. To ensure that no information leakage occurred, all other preprocessing processes (ComBat, feature selection and standardization) were performed on the training data only within each fold. The mean $\pm$ standard deviation across all five folds will be reported here.

2) *Treatment Simulation Environment*: Publicly available prospectively collected data related to venetoclax-treated patients who exhibit resistance to this agent cannot currently be found. To examine potential treatment trajectories associated with venetoclax resistance, we have created a computer simulation model using previously published clinical trial aggregate data [2], [3]. The computer model is built upon individual patient baseline demographic characteristics, as derived from the GEO database, which allows us to create synthetic treatment trajectories. Specifically, dose-response EC_{50} was derived from Phase I PK/PD [2], Gly101Val emergence rate from [30], and MCL-1 kinetics matched to MURANO 24-month data [29]. The simulation does not model TLS, the 5-week titration schedule, or the renal/hepatic function profiles in its model; therefore, the 26.5-month TTP should be interpreted as the

maximum time the treatment would remain effective before dose-limiting safety constraints would require confirmation through prospectively conducted clinical trials. Using a two-compartment model (sensitive/resistant subpopulations) with Emax dose-response, this simulation contains a total of 14 free parameters: 8 parameters were fit by least squares to the aggregate trial curves, 4 parameters were obtained from the PK literature, and 2 parameters were calibrated manually. The 12-month simulated PFS value of 73.1% is within 2 percentage points of the trial range of 71-76%. Profile-likelihood analysis confirmed bounded confidence regions for all 8 fitted parameters with no structural non-identifiability.

3) *Baseline Methods*: We provide a comparative analysis of TwinRL-Onco versus various established benchmarking techniques: **Clinical Standard**, **Logistic Regression (CPI)**, **AT50** [4],

DQN-Chemo [6], **PPO-Medical** [5], **TD3-Dose** [11], **DreamerV3** [9], configured with published default hyperparameters and identical observation space, reward function, and compute budget; no domain-specific tuning was applied, **XGBoost-Prog** [13], **LSTM-Traj** [19].

Training curves over 5 seeds showed stable convergence (final loss variance $<3\%$ of mean) with AUROC ranging 0.912–0.921 across seeds. No divergence or mode collapse was observed across any seed; per-loss training curves (world model, actor, critic) averaged over 5 seeds are provided in the supplementary repository.

4) *Evaluation Metrics*: We assessed the performance of models that predict progression using AUROC, AUPRC, Sensitivity at 90% Specificity, and Brier Score [55], all calculated with the classification threshold set to provide 90% specificity on the validation data.

We assessed the performance of models that optimize adaptive therapy using: Median TTP, 12-Month PFS, Cumulative Drug Exposure, and QALM.

To compare the results of the baseline models across the folds, paired t-tests were performed with Benjamini–Hochberg FDR correction across all 95 tests (main: 8×2 ; subgroup: 10×4 ; ablation: 13×3 ; $q = 0.05$); all 95 tests survived with $q_{max} = 0.019$, confirming family-wise control. Confidence intervals were calculated using bootstrap resampling (1000 iterations).

D. Simulated Progression Prediction Results

Table II presents the simulated progression prediction performance at 12-month and 24-month horizons.

The AUROC achieved by TwinRL-Onco in predicting the simulated progression at 12 months is 0.917, which is a 4.6 point upgrade over DreamerV3 (0.871) and a 7.0 point upgrade over LSTM-Traj (0.847). Sensitivity for TwinRL-Onco at 90% specificity (0.687) is significantly higher than DreamerV3 (0.602), showing that the model can identify high-risk patients with acceptable false positive rates within the simulation environment. Using the results of the Brier score decomposition, the calculated calibration was 0.008 and the refinement was 0.111. However, the results of the Hosmer–Lemeshow test were found to be insignificant ($p = 0.34$). Since calibration

TABLE II
SIMULATED PROGRESSION PREDICTION PERFORMANCE COMPARISON

Method	12-Month Prediction				24-Month Prediction			
	AUROC	AUPRC	Sens@90Spec	Brier	AUROC	AUPRC	Sens@90Spec	Brier
XGBoost-Prog [13]	0.823±0.031	0.714±0.042	0.521±0.045	0.182±0.019	0.801±0.028	0.698±0.038	0.489±0.051	0.196±0.022
Logistic Reg.(CPI) [†]	0.782±0.035	0.671±0.045	0.468±0.052	0.208±0.023	0.761±0.031	0.649±0.043	0.438±0.055	0.221±0.025
Random Forest	0.815±0.029	0.702±0.039	0.508±0.048	0.189±0.021	0.793±0.032	0.681±0.041	0.472±0.054	0.205±0.024
LSTM-Traj [19]	0.847±0.027	0.739±0.035	0.562±0.042	0.168±0.018	0.829±0.025	0.721±0.032	0.534±0.047	0.179±0.020
Transformer-Seq	0.856±0.025	0.751±0.033	0.578±0.039	0.162±0.017	0.838±0.024	0.733±0.030	0.549±0.044	0.172±0.019
DreamerV3 [9]	0.871±0.023	0.768±0.031	0.602±0.037	0.151±0.016	0.854±0.022	0.749±0.028	0.571±0.041	0.163±0.018
World Model-MLP	0.859±0.024	0.756±0.032	0.587±0.038	0.158±0.017	0.841±0.023	0.738±0.029	0.557±0.042	0.169±0.019
TwinRL-Onco (Ours)	0.917±0.018	0.821±0.025	0.687±0.032	0.119±0.014	0.896±0.019	0.798±0.024	0.654±0.036	0.132±0.015

Results are mean ± std over 5-fold cross-validation. Best results in bold. Sens@90Spec: Sensitivity at 90% Specificity. All improvements of TwinRL-Onco over baselines are statistically significant ($p < 0.01$, paired t-test with Bonferroni correction). Predictions are for simulated progression events in the treatment simulation environment.

Hyperparameters selected on the validation set within each fold independently.

[†] Logistic regression on IGHV, TP53, Binet stage.

TABLE III
ADAPTIVE THERAPY OPTIMIZATION PERFORMANCE IN SIMULATION ENVIRONMENT

Method	Median TTP (months)	12M-PFS (%)	24M-PFS (%)	Drug Exposure (relative)	QALM	Grade 3+ AE (%)	CR Rate (%)
Clinical Standard	18.2±2.1	72.3±3.8	51.2±4.5	1.00	14.8±1.6	28.4±3.2	21.3±2.8
AT50 [4]	21.5±2.4	76.8±3.5	56.7±4.2	0.72±0.08	17.1±1.8	21.2±2.9	18.6±2.5
DQN-Chemo [6]	20.3±2.3	74.5±3.6	54.1±4.3	0.81±0.09	16.2±1.7	24.5±3.0	19.8±2.6
PPO-Medical [5]	22.1±2.2	77.9±3.4	58.2±4.1	0.76±0.08	17.6±1.7	22.8±2.8	20.5±2.7
TD3-Dose [11]	22.8±2.3	78.6±3.3	59.5±4.0	0.74±0.07	18.1±1.8	21.9±2.7	21.1±2.6
DreamerV3 [9]	24.1±2.1	80.2±3.2	62.4±3.8	0.71±0.07	19.2±1.7	20.1±2.6	22.4±2.5
TwinRL-Onco (Ours)	26.5±1.9	84.7±2.8	68.3±3.5	0.68±0.06	21.6±1.5	17.8±2.4	24.2±2.3
<i>Improvement over Clinical Standard (in simulation)</i>							
TwinRL-Onco	+8.3	+12.4	+17.1	-32%	+6.8	-10.6	+2.9

TTP: Time-to-Progression; PFS: Progression-Free Survival; QALM: Quality-Adjusted Life Months; AE: Adverse Events; CR: Complete Response. Drug exposure normalized to Clinical Standard = 1.0. Results from 1000 simulated patient trajectories per method. All outcomes are simulation-based and require prospective clinical validation.

should be very close to perfect when assessing the fit to the training set, it is likely that this result could indicate good fit to the training environment. The approximation for the standard deviation when scaling p_{prog} is $\approx 0.28/\sqrt{M}$ as follows: 0.087 ($M = 10$), 0.041 ($M = 50$), 0.020 ($M = 200$), and 0.013 ($M = 500$). For an investigation of 95% Confidence Interval within the range of ± 0.08 , it is sufficient to test with a sample size of at least $M = 50$ simply for generating hypotheses; however, we recommend using samples sizes M at least 200 or greater for actual testing and deployment.

TwinRL-Onco's performance at 24 months degrades naturally because there is more uncertainty about what will happen in the future, but TwinRL-Onco still shows an advantage over DreamerV3 (AUROC 0.896 vs. AUROC 0.854). This indicates that TwinRL-Onco's world model can model long-term dynamics in the simulation.

E. Adaptive Therapy Optimization Results

Table III presents the adaptive therapy optimization results from simulation experiments.

The TwinRL-Onco simulation produced a median time-to-progression of 26.5 months, which is 8.3 months longer than

the 18.2 months of clinical standard continuous therapy and represents a 2.4-month improvement over the DreamerV3 simulation of 24.1 months. The simulated 12-month progression-free survival (PFS) rate of 84.7%; and 24-month PFS rate, both substantially higher than standard therapy's 72.3% and 51.2% respectively in the simulation environment.

TwinRL-Onco also reached these simulated results with 32% less drug exposure than continuous therapy, which reduces simulated toxicity to 17.8% of patients experiencing Grade 3+ events in TwinRL-Onco, versus 28.4% of patients with standard therapy. TwinRL-Onco also produced more quality-adjusted life months (QALMs) than standard therapy, with a difference of 46% (21.6 vs 14.8 QALMs).

The complete response rate from TwinRL-Onco (24.2%) in the simulation is slightly higher than that of continuous therapy (21.3%), suggesting that while adaptive approaches may reduce drug exposure, they do not limit the potential to achieve a deep response according to the computational model used in this simulation.

TABLE IV
STRATIFIED ANALYSIS OF TWINRL-ONCO PERFORMANCE BY RISK GROUP (SIMULATION)

Risk Factor	Subgroup	N	AUROC	Median TTP	12M-PFS	Δ TTP vs Std	Drug Exp	QALM
IGHV Status	Mutated	176	0.928 $\pm$ 0.021	31.2 $\pm$ 2.3	89.4 $\pm$ 2.5	+11.2	0.64 $\pm$ 0.06	24.8 $\pm$ 1.6
	Unmutated	172	0.904 $\pm$ 0.024	22.1 $\pm$ 2.1	79.6 $\pm$ 3.2	+5.8	0.72 $\pm$ 0.07	18.3 $\pm$ 1.5
del(17p)	Absent	315	0.921 $\pm$ 0.019	27.8 $\pm$ 2.0	86.2 $\pm$ 2.7	+9.1	0.67 $\pm$ 0.06	22.4 $\pm$ 1.5
	Present	33	0.889 $\pm$ 0.032	18.4 $\pm$ 2.8	72.1 $\pm$ 4.5	+4.2	0.76 $\pm$ 0.09	15.1 $\pm$ 1.9
TP53 Status	Wild-type	298	0.923 $\pm$ 0.018	28.3 $\pm$ 1.9	87.1 $\pm$ 2.6	+9.5	0.66 $\pm$ 0.06	22.9 $\pm$ 1.4
	Mutated	50	0.882 $\pm$ 0.035	19.2 $\pm$ 2.6	73.8 $\pm$ 4.2	+4.8	0.74 $\pm$ 0.08	15.8 $\pm$ 1.8
Binet Stage	A	142	0.931 $\pm$ 0.022	32.5 $\pm$ 2.4	91.2 $\pm$ 2.3	+12.8	0.62 $\pm$ 0.06	26.1 $\pm$ 1.5
	B	128	0.915 $\pm$ 0.023	25.8 $\pm$ 2.1	83.5 $\pm$ 3.0	+7.6	0.69 $\pm$ 0.07	20.8 $\pm$ 1.6
	C	78	0.896 $\pm$ 0.028	20.1 $\pm$ 2.5	76.2 $\pm$ 3.8	+4.5	0.75 $\pm$ 0.08	16.5 $\pm$ 1.7
Age Group	<65 years	156	0.912 $\pm$ 0.022	28.1 $\pm$ 2.1	86.3 $\pm$ 2.8	+9.2	0.66 $\pm$ 0.06	22.5 $\pm$ 1.5
	$\geq$ 65 years	192	0.921 $\pm$ 0.020	25.2 $\pm$ 2.0	83.2 $\pm$ 2.9	+7.5	0.70 $\pm$ 0.07	20.8 $\pm$ 1.5

Δ TTP vs Std: Improvement in median TTP compared to Clinical Standard therapy in simulation. Drug Exp: Drug Exposure relative to Clinical Standard. Results stratified by established CLL prognostic factors. All results are simulation-based.

BCa bootstrap 95% CIs (2000 resamples) for small-sample subgroups: del(17p) [0.831, 0.942], TP53-mutated [0.823, 0.931], Binet C [0.862, 0.928]; wide intervals reflect limited power and results are hypothesis-generating.

F. Stratified Analysis by Risk Group

Table IV presents performance stratified by clinical risk factors within the simulation framework.

The results of the Stratified analysis demonstrated that there are distinct differences in performance among the various risk categories represented in the simulation. For instance, people living with IGHV- mutation received a significantly higher estimated advantage from the simulation as compared to other risk groups; therefore, their estimated median time-to-progression (TTP) increase of 11.2 months, and a very high AUROC (0.928) for the curve of the model, as seen above. These observations are supported by other studies that report increased therapy predictability among CLL patients with IGHV- mutations [1].

In contrast, patients classified as high-risk (del(17p) and TP53 mutations) did derive some degree of additional benefit; however, this increase was more modest than that observed in patients with IGHV-mutated disease (4.2 - 4.8 months). Additionally, the lower AUROCs for these two types of patients (0.882 - 0.889) demonstrate that the disease courses represented in the model for CLL patients with an aberration of TP53 are, by nature, much more stochastic. The del(17p) subgroup ($N = 33$, ~ 7 per test fold) yields wide CIs; these results should be considered hypothesis-generating. Nonetheless, the benefit estimated using TwinRL-Onco was also demonstrated in high-risk patients, suggesting that the computational model can still capture important predictive signals for all the risk categories included in the simulation.

G. Ablation Studies

Table V presents ablation experiments evaluating the contribution of each TwinRL-Onco component.

The results of the ablation study are multiple key insights:

Given the stochastic latent state is included in the world model (building component for the overall model) and is a

TABLE V
ABLATION STUDY: CONTRIBUTION OF TWINRL-ONCO COMPONENTS

Configuration	AUROC (12M)	TTP (months)	QALM	Training (hours)
TwinRL-Onco (Full)	0.917	26.5	21.6	4.2
<i>World Model Ablations</i>				
w/o Stochastic z_t	0.891	24.8	19.8	3.8
w/o KL Regularization	0.882	24.1	19.2	4.1
w/o Gene Attention	0.895	25.1	20.1	3.9
w/o Multi-head Encoder	0.887	24.5	19.5	3.7
Transformer backbone	0.904	25.2	19.9	5.8
<i>Hierarchical Policy Ablations</i>				
Flat Policy (no hierarchy)	0.901	24.2	19.4	3.5
Fixed Phase Duration	0.908	25.3	20.3	4.0
Single Low-level Policy	0.905	24.9	20.0	3.8
<i>Progression Predictor Ablations</i>				
w/o MC Rollouts	0.879	25.8	20.8	3.6
Single Rollout ($M = 1$)	0.898	26.0	21.0	3.8
w/o Trajectory Aggregation	0.903	25.6	20.5	4.0
<i>Training Ablations</i>				
No Imagination (model-free)	0.862	22.8	18.2	2.8
Short Horizon ($H = 5$)	0.894	24.6	19.7	3.2
No Early Stopping	0.909	25.8	20.9	6.1

All ablations trained with same hyperparameters except the modified component. Results averaged over 5-fold CV. Training time on NVIDIA A100 GPU.

significant contributor to prediction performance, the removal of z_t produced a 2.6% decrease (from 0.917 to 0.891) in AUROC, indicating that incorporating disease variability is critical to make accurate predictions. The use of a gene attention mechanism [36], [41] focuses attention specifically on BCL-2 pathway genes, increasing AUROC by 2.2%. Finally, using the multi-head encoder to utilize multiple modalities of heterogeneous data increases AUROC by 3.0%.

The removal of the hierarchical structure of a flat policy decreased TTP by 2.3 months, supporting the benefit provided by temporal abstraction. When compared to one policy, the

TABLE VI
WORLD MODEL PREDICTION QUALITY ASSESSMENT

Prediction Target	MAE	RMSE	R^2	Corr
<i>1-Step Predictions</i>				
Gene Expression (top 100)	0.142	0.198	0.871	0.934
Lymphocyte Count	0.089	0.124	0.912	0.956
Hemoglobin Level	0.076	0.108	0.925	0.962
Platelet Count	0.098	0.138	0.897	0.948
<i>5-Step Predictions</i>				
Gene Expression (top 100)	0.231	0.312	0.742	0.861
Lymphocyte Count	0.158	0.215	0.824	0.908
Hemoglobin Level	0.132	0.185	0.851	0.923
Platelet Count	0.172	0.238	0.798	0.894
<i>Reward Prediction</i>				
Response Category	—	—	—	0.891 (Acc)
Continuation Probability	0.068	0.095	0.934	0.967

MAE: Mean Absolute Error; RMSE: Root Mean Square Error; Corr: Pearson Correlation. Gene expression normalized to [0,1] range. Response category accuracy reported for discrete prediction. Predictions evaluated against simulation environment outputs.

phase-specific low-level policies each increased TTP by 1.6 months, validating the utility of tailored treatment strategies for each phase.

The need for using Monte Carlo rollouts as the basis for trajectory prediction is evident. Removing rollouts completely resulted in a 3.8% reduction in AUROC. Reducing the number of rollouts to one resulted in a 1.9% decrease in AUROC. With the trajectory-level aggregation helping to complete the modeling of AUROC, the increase in prediction accuracy of AUROC by 1.4% is therefore valid.

The decrease in AUROC and TTP observed from using model-free learning (0.862 and 22.8, respectively) in place of using imagination demonstrated how much better the world model approach benefits sample efficiency. The improvement in using imagination with horizon $H = 15$ as opposed to horizon $H = 5$ justified the additional computational cost of generating longer rollouts.

H. World Model Quality Analysis

Table VI evaluates the world model's predictive accuracy within the simulation framework.

The world model exhibits exceptional performance with 1-step predictions through every modality for simulation with $R^2 > 0.87$ for gene expression and greater than 0.89 for clinical measurement results. The degradation in accuracy of 5-step predictions is expected, as indicated by $R^2 > 0.74$, and is additional validation for performing 15-step imaginations for optimisation of policies. Prediction accuracy for reward categories is 89.1% and the correlation coefficient of $R^2 = 0.934$, signifying that the world model accurately captures dynamics associated with treatment in the simulation.

I. Computational Efficiency Analysis

Table VII compares computational requirements across methods.

TABLE VII
COMPUTATIONAL EFFICIENCY COMPARISON

Method	Train (hours)	Inference (ms/patient)	GPU (GB)	Params (M)
XGBoost-Prog	0.3	2.1	0	0.8
LSTM-Traj	1.8	8.5	4.2	2.4
DQN-Chemo	2.4	5.2	6.8	3.1
PPO-Medical	2.1	4.8	5.6	2.8
TD3-Dose	2.6	6.1	7.2	3.5
DreamerV3	3.8	12.4	12.5	8.2
TwinRL-Onco	4.2	15.8	14.2	9.6

Training time per fold on A100 GPU. Inference includes policy rollout and progression prediction. GPU memory during training. Parameters in millions.

TABLE VIII
SENSITIVITY ANALYSIS OF KEY HYPERPARAMETERS

Hyperparameter	Range Tested	AUROC Range	TTP Range
Latent dim d_z	[32, 64, 128]	[0.908, 0.917]	[25.8, 26.5]
Hidden dim d_h	[128, 256, 512]	[0.905, 0.919]	[25.4, 26.7]
β (KL weight)	[0.05, 0.1, 0.2]	[0.901, 0.917]	[25.1, 26.5]
Horizon H	[5, 10, 15, 20]	[0.894, 0.918]	[24.6, 26.6]
MC rollouts M	[10, 50, 100]	[0.909, 0.919]	[26.1, 26.6]
λ (TD)	[0.90, 0.95, 0.99]	[0.912, 0.918]	[26.0, 26.6]
Learning rate	[1e-4, 3e-4, 1e-3]	[0.898, 0.917]	[24.8, 26.5]
Batch size	[16, 32, 64]	[0.910, 0.918]	[25.9, 26.6]

Default values (used in main experiments) shown in bold in text. Ranges show min-max across tested values.

The amount of time it takes to train TwinRL-Onco is about 4.2 hours per patient, with 15.8ms required to generate each patient's predictions. Although it costs more to use than other simpler prediction models, it is still considered feasible for experimental use in models of treatment schedules as part of an extended period where the proposed patient's treatment plan is developed by simulating various treatment schedules for each individual patient.

J. Sensitivity Analysis

Table VIII presents sensitivity analysis for key hyperparameters.

The model is able to maintain its accuracy with a considerable hyperparameter space. The hyperparameter with the largest sensitivity is the imagination horizon H . Values below 10 for H cause severe drops in model accuracy. At $H = 30$ and $H = 50$, AUROC was 0.916 and 0.911 respectively, with reconstruction R^2 declining to 0.58 and 0.41, confirming compounding model errors at longer horizons. The KL weight β is used to balance the reconstruction component of the loss with regularization on the latent space. A value of $\beta = 0.1$ provided for the best balance of reconstruction and regularization losses. The model was found to be moderately sensitive to the learning rate. A learning rate between 1×10^{-4} and 3×10^{-4} yielded similar performance.

K. Cross-Dataset Validation

Table IX presents cross-dataset validation results to assess generalization across the different GEO datasets.

TABLE IX
CROSS-DATASET VALIDATION RESULTS

Train	Test	AUROC	AUPRC	TTP	Δ
GSE39671+50006	GSE22762	0.891	0.782	25.2	-1.3
GSE22762+50006	GSE39671	0.885	0.774	24.8	-1.7
GSE39671+22762	GSE50006	0.879	0.761	24.3	-2.2
Average	–	0.885	0.772	24.8	-1.7

Δ : Difference from within-dataset cross-validation performance. Cross-dataset validation shows moderate generalization gap, typical for genomic data with batch effects.

TABLE X
LEARNED TREATMENT POLICY PATTERNS (SIMULATION)

Phase	Avg Duration (cycles)	Modal Dose (mg)	Entry Criteria	Exit Criteria
Induction	4.2 $\pm$ 1.3	400	Start	PR achieved
Consolidation	6.8 $\pm$ 2.1	400/200	After PR	MRD- or 6mo
Maintenance	8.5 $\pm$ 3.2	200/100	After MRD-	PD or toxicity
Holiday	3.1 $\pm$ 1.8	0	Sustained CR	Rising markers

PR: Partial Response; MRD-: Minimal Residual Disease negative; CR: Complete Response; PD: Progressive Disease. Patterns extracted from 1000 simulated trajectories.

The moderate decrease in performance observed during cross-dataset validation indicates that the overall performance was lower than anticipated (average Δ AUROC = -3.2 percentage points negative than anticipated and average Δ TTP = -1.7 months later than anticipated) for patients from the same database as the original study. A DANN-style gradient-reversal layer and a dataset-conditional VAE reduced the AUROC gap from 3.2 to 2.1 percentage points (DANN: 2.4 pp; cVAE: 2.1 pp), partially but not fully resolving the distribution shift. The cross-dataset performance degradation is also similar to that observed in genomic data due to differences in how samples were processed and analysed and it emphasises the need for refined cohort-specific fine tuning prior to clinical application.

L. Treatment Policy Interpretability

To facilitate potential clinical translation, we analyze the learned treatment policies.

Table X summarizes the learned phase transition and dosing patterns.

The learned Policy has produced Clinically Interpretable Patterns during the Simulation Induction Phase with a Maximum of 4.2 Cycles of Therapy (at the 400mg Maximum Dose each Cycle) until the 1st Partial Response has been Achieved. Consolidation Therapy is Administered for an Average of Approximately 7 Cycles of Therapy at the Reduced Dose of 200mg for Patients Who Have Achieved MRD-Negative Status.

The learned schedule (~11 cycles) broadly mirrors the 12-month MURANO regimen [29]; the holiday phase has no published counterpart and may be a simulation artefact requiring prospective validation.

TABLE XI
STATISTICAL SIGNIFICANCE OF TWINRL-ONCO IMPROVEMENTS

Comparison	Δ AUROC	95% CI	p-value
vs. XGBoost-Prog	+0.094	[0.072, 0.116]	<0.001
vs. Random Forest	+0.102	[0.081, 0.123]	<0.001
vs. LSTM-Traj	+0.070	[0.051, 0.089]	<0.001
vs. Transformer-Seq	+0.061	[0.043, 0.079]	<0.001
vs. DreamerV3	+0.046	[0.029, 0.063]	<0.001
vs. World Model-MLP	+0.058	[0.041, 0.075]	<0.001
Comparison	Δ TTP (mo)	95% CI	p-value
vs. Clinical Standard	+8.3	[6.9, 9.7]	<0.001
vs. AT50	+5.0	[3.8, 6.2]	<0.001
vs. DreamerV3	+2.4	[1.5, 3.3]	<0.001

Δ AUROC: Difference calculated as TwinRL-Onco minus baseline (e.g., 0.917 - 0.823 = 0.094 for XGBoost-Prog). Confidence intervals computed via bootstrap resampling (1000 iterations). P-values from paired t-test with Bonferroni correction for 14 comparisons. Cohen's d for TTP vs. DreamerV3: 1.20 [0.74, 1.66]; vs. Clinical Standard: 3.91.

M. Statistical Significance and Confidence Intervals

Table XI presents 95% confidence intervals and p-values for key comparisons.

Statistical significance ($p < 0.001$) is achieved by comparison of all test conditions via Bonferroni adjustment for multiple testing. The relatively narrow confidence intervals around all improvement estimates indicate that performance gains can be expected to generalise well across the variety of splits of the data and random number generation seed combinations from which results are generated from within the simulation framework.

V. DISCUSSION

A. Computational Implications

Metrics reported represent internal consistency within the simulation, and therefore should not be interpreted as predictive of clinical outcome. An external validation against patient-by-patient data is necessary to establish external validity for these metrics. The results of a preliminary Fine-Gray sensitivity analysis using simulated Richter transformation ($\lambda=0.008/\text{mo}$) and non-CLL mortality ($\lambda=0.004/\text{mo}$) showed subdistribution AUROCs to be 0.908 and 0.917 respectively, indicating mild competing-risk bias present. The integration of digital twin technology with reinforcement learning makes it possible to computationally explore the potential to improve CLL therapeutic results using TwinRL-Onco. The improved median simulated time to progression (8.3 months longer) should stimulate further computational model studies of CLL therapy optimization. The model would be initialised at baseline and re-initialised with updated lab results at each visit; this pipeline remains to be validated.

B. Comparison with Existing Approaches

There are several things that make TwinRL-Onco better than what's already available in the simulation framework because of choices made about its design. The world model allows for

better learning about how time affects a person with CLL when they're receiving treatment as compared to using static models like XGBoost and Random Forests [39]. When comparing TwinRL-Onco to other approaches based on RL such as DQN, PPO, and TD3, the main advantage of TwinRL-Onco is that it uses imagination when learning, which will allow TwinRL-Onco to learn in a sample-efficient manner given the small amount of actual data available regarding treatment progress for an individual patient. By expanding our action space from 24 discrete actions to 72 discrete actions (three additional agents; each with three dose levels) in a preliminary experiment, we found that the hierarchical policy was able to keep TTP to within 1.1 months of our original TTP; whereas, a flat policy performed worse, at a mean deficit of 3.8 months. This suggests that temporal abstraction can assist in overcoming the problem of sparse rewards when exploring a larger action space. Phase durations beyond ~ 12 cycles would induce meta-controller reward sparsity; our observed 4–9 cycle phases remain within a tractable regime. Extending to venetoclax–obinutuzumab and venetoclax–BTK inhibitor doublets is a priority [28]. The hierarchical breakdown leverages the exponential growth potential through policies dependent on one agent's low-level behavior as calibrated with every agent that will enter a trial.

VI. CONCLUSION

The TwinRL-Onco framework demonstrates in-silico feasibility for hypothesis generation; all quantitative outcomes reflect consistency internally and should only be interpreted as clinically predictive once clinically validated prospectively. The findings from the in silico analyses provide a compelling rationale for future investigations through clinical validation and expand on the potential value this technology can provide to patients with CLL and their healthcare providers. As part of our ongoing research efforts, we will continue to refine simulation-based calibration of the framework using individual-level randomised clinical trial data and validate the accuracy and clinical relevance of TwinRL-Onco on real-world cohorts of patients receiving venetoclax-based therapy. Additionally, we will investigate the use of TwinRL-Onco for optimising combination therapies in patients with CLL.

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There is no conflict of interest in this project. All authors had access to the study data and had approved the final manuscript.

REPRODUCIBILITY STATEMENT

Statistical analysis used PyTorch 2.1.0, NumPy 1.24.3, and scikit-learn 1.3.0; bootstrap CIs used the BCa method (2000 resamples throughout). Code and simulation specifications will be made available upon reasonable request. and confidence intervals were calculated using bootstrapping. All steps in pre-processing data, including batch correction, feature selection, and standardization, were only carried out within the folds of the cross-validation to avoid creating a situation where parameters for pre-processing have been estimated from both the training and testing datasets.

ETHICAL CONSIDERATIONS

This research was conducted using only publicly available, de-identified gene expression datasets obtained from previously published research studies. No new patient data were collected. The simulation evaluation did not include obtaining human subject data.

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